



The Greenland Ice Sheet

The largest glacier in Earth's Northern Hemisphere, covering 80 percent of the island of Greenland and holding a significant fraction of the world's freshwater

The Greenland ice sheet is the world's second largest ice sheet (and second largest glacier) after the Antarctic Ice sheet (see pp. 306–307), covering 660,000 square miles (1.71 million square km) of Greenland. It has an estimated volume of 680,000 cubic miles (2.85 million cubic km), with an average thickness of about 5,480 ft (1,670 m) and a maximum thickness of 10,515 ft (3,205 m). Some of its ice is thought to be up to 110,000 years old.

Shape and movement

The ice sheet's surface is slightly domed, reaching a height of about 10,800 ft (3,290 m) above sea level at the top of the dome. From here, ice moves slowly down toward the ice sheet's edges, where it is mostly constrained by coastal mountains; it only reaches the sea along a broad front in a few places. As a result, Greenland (unlike Antarctica) has no ice-shelves. In many

places, the ice flows through gaps in the mountains in the form of outlet glaciers. When these reach the coast, they discharge enormous numbers of icebergs into the sea. One outlet glacier, the Jakobshavn Glacier in western Greenland, is the world's fastest-flowing glacier. At its sea end, the ice flows at about 3 ft (1 m) per hour.

Threats to sea level and freshwater

The Greenland ice sheet is melting due to global warming. If it melted completely, the world's oceans would rise by about 24 ft (7.2 m), drowning many of the world's major cities and leading to loss of about 6.5 percent of the world's freshwater reserves.

◀ ARCTIC PREDATOR

Although rarely venturing onto the ice sheet itself, polar bears are active around its edges on the Greenland coast, where they hunt and feed on seals.



△ ICEBERG MAKER

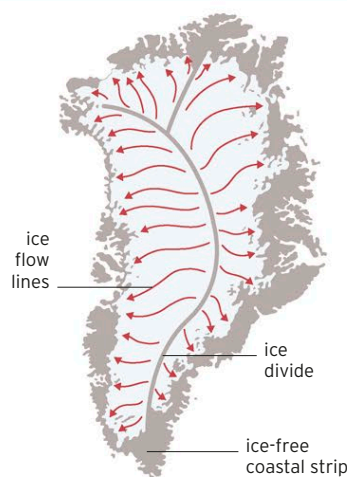
A helicopter flies past the end of Graah Glacier, an outlet glacier that flows out of the Greenland ice sheet. From this wall of ice, huge icebergs break off, or calve, from time to time.

◀ WASHED OUT

From the snout of Russell Glacier in central western Greenland emerges a river of meltwater, skirting rock debris that has been carried there by the ice.

ICE MOVEMENT AND ICE DIVIDES

Although ice covers most of Greenland, there is an ice-free zone of variable width around much of the island's coast. Snow falling on the ice sheet compacts to ice, which then slowly flows down in the direction of the coast along the flow lines shown here. Imaginary lines called ice divides separate regions of ice moving toward different coasts.



The Jakobshavn Glacier produces about **99 million tons (90 million tonnes)** of icebergs every day